

Multiple Choice (10 marks)

1. Which of the following is a primary standard for use in standardizing bases?
 - (a) Ammonium hydroxide
 - (b) Sulfuric acid
 - (c) Acetic acid
 - (d) Potassium hydrogen phthalate
2. Which of the following experimental observations were explained by Planck's quantum theory?
 - (a) Blackbody radiation curves
 - (b) Emission spectra of diatomic molecules
 - (c) Electron diffraction patterns
 - (d) Temperature dependence of reaction rates
3. The anhydride of $\text{Ba}(\text{OH})_2$ is
 - (a) BaH_2
 - (b) BaOH
 - (c) Ba
 - (d) BaO
4. The T_2 of an NMR line is 15 ms. Ignoring other sources of linebroadening, calculate its linewidth.
 - (a) 0.0471 Hz
 - (b) 21.2 Hz
 - (c) 42.4 Hz
 - (d) 66.7 Hz
5. Which of the following metal ions cannot be used as a paramagnetic quencher?
 - (a) Ti^{3+}
 - (b) Cr^{3+}
 - (c) Fe^{3+}
 - (d) Zn^{2+}

True / False (10 marks)

Answer True or False

6. True or False: H_3O^+ and H_2O form a conjugate acid-base pair in acid-base chemistry.
7. True or False: BeCl_2 is less likely to act as a Lewis acid compared to ZnCl_2 because it has a higher oxidation state.
8. True or False: The solid-state structures of the principal allotropes of elemental boron are composed of B_{12} icosahedra.
9. True or False: The ^{13}C spectrum of hexane shows five distinct chemical shifts because it has multiple isomers.
10. True or False: The limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule is $2.5 R$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the trends in atomic radii of lanthanide elements from lanthanum to lutetium, and evaluate which statement regarding these trends is misleading or incorrect.
12. Discuss the relationship between spontaneity and entropy in thermodynamic processes, explaining why the total entropy of the system and surroundings must increase for a process to be spontaneous.

End of Paper